What is a data lake?

A data lake is a central location that holds a large amount of data in its native, raw format. Compared to a hierarchical data warehouse, which stores data in files or folders, a data lake uses a flat architecture and object storage to store the data. Object storage stores data with metadata tags and a unique identifier, which makes it easier to locate and retrieve data across regions, and improves performance. By leveraging inexpensive object storage and open formats, data lakes enable many applications to take advantage of the data.

Data lakes were developed in response to the limitations of data warehouses. While data warehouses provide businesses with highly performant and scalable analytics, they are expensive and proprietary and can't handle the modern use cases most companies are looking to address. Data lakes are often used to consolidate all of an organization's data in a single, central location, where it can be saved "as is," without the need to impose a schema (i.e., a formal structure for how the data is organized) up front like a data warehouse does. Data in all stages of the refinement process can be stored in a data lake: raw data can be ingested and stored right alongside an organization's structured, tabular data sources (like database tables), as well as intermediate data tables generated in the process of refining raw data. Unlike most databases and data warehouses, data lakes can process all data types — including unstructured and semi-structured data like images, video, audio and documents — which are critical for today's machine learning and advanced analytics use cases.

Why would you use a data lake?

First and foremost, data lakes are open format, so users avoid lock-in to a proprietary system like a data warehouse, which has become increasingly important in modern data architectures. Data lakes are also highly durable and low cost, because of their ability to scale and leverage object storage. Additionally, advanced analytics and machine learning on unstructured data are some of the most strategic priorities for enterprises today. The unique ability to ingest raw data in a variety of formats (structured, unstructured, semi-structured), along with the other benefits mentioned, makes a data lake the clear choice for data storage.

When properly architected, data lakes enable the ability to:

Power data science and machine learning

Data lakes allow you to transform raw data into structured data that is ready for SQL analytics, data science and machine learning with low latency. Raw data can be retained indefinitely at low cost for future use in machine learning and analytics.

Centralize, consolidate and catalogue your data

A centralized data lake eliminates problems with data silos (like data duplication, multiple security policies and difficulty with collaboration), offering downstream users a single place to look for all sources of data.

Quickly and seamlessly integrate diverse data sources and formats

Any and all data types can be collected and retained indefinitely in a data lake, including batch and streaming data, video, image, binary files and more. And since the data lake provides a landing zone for new data, it is always up to date.

Democratize your data by offering users self-service tools

Data lakes are incredibly flexible, enabling users with completely different skills, tools and languages to perform different analytics tasks all at once.

Data lake challenges

Despite their pros, many of the promises of data lakes have not been realized due to the lack of some critical features: no support for transactions, no enforcement of data quality or governance, and poor performance optimizations. As a result, most of the data lakes in the enterprise have become data swamps.

Reliability issues

Without the proper tools in place, data lakes can suffer from data reliability issues that make it difficult for data scientists and analysts to reason about the data. These issues can stem from difficulty combining batch and streaming data, data corruption and other factors.

Slow performance

As the size of the data in a data lake increases, the performance of traditional query engines has traditionally gotten slower. Some of the bottlenecks include metadata management, improper data partitioning and others.

Lack of security features

Data lakes are hard to properly secure and govern due to the lack of visibility and ability to delete or update data. These limitations make it very difficult to meet the requirements of regulatory bodies.

For these reasons, a traditional data lake on its own is not sufficient to meet the needs of businesses looking to innovate, which is why businesses often operate in complex architectures, with data siloed away in different storage systems: data warehouses, databases and other storage systems across the enterprise. Simplifying that architecture by unifying all your data in a data lake is the first step for companies that aspire to harness the power of machine learning and data analytics to win in the next decade.

How a lakehouse solves those challenges

The answer to the challenges of data lakes is the lakehouse, which adds a transactional storage layer on top. A lakehouse that uses similar data structures and data management features as those in a data warehouse but instead runs them directly on cloud data lakes. Ultimately, a lakehouse allows traditional analytics, data science and machine learning to coexist in the same system, all in an open format.

A lakehouse enables a wide range of new use cases for cross-functional enterprise-scale analytics, BI and machine learning projects that can unlock massive business value. Data analysts can harvest rich insights by querying the data lake using SQL, data scientists can join and enrich data sets to generate ML models with ever greater accuracy, data engineers can build automated ETL pipelines, and business intelligence analysts can create visual dashboards and reporting tools faster and easier than before. These use cases can all be performed on the data lake simultaneously, without lifting and shifting the data, even while new data is streaming in.

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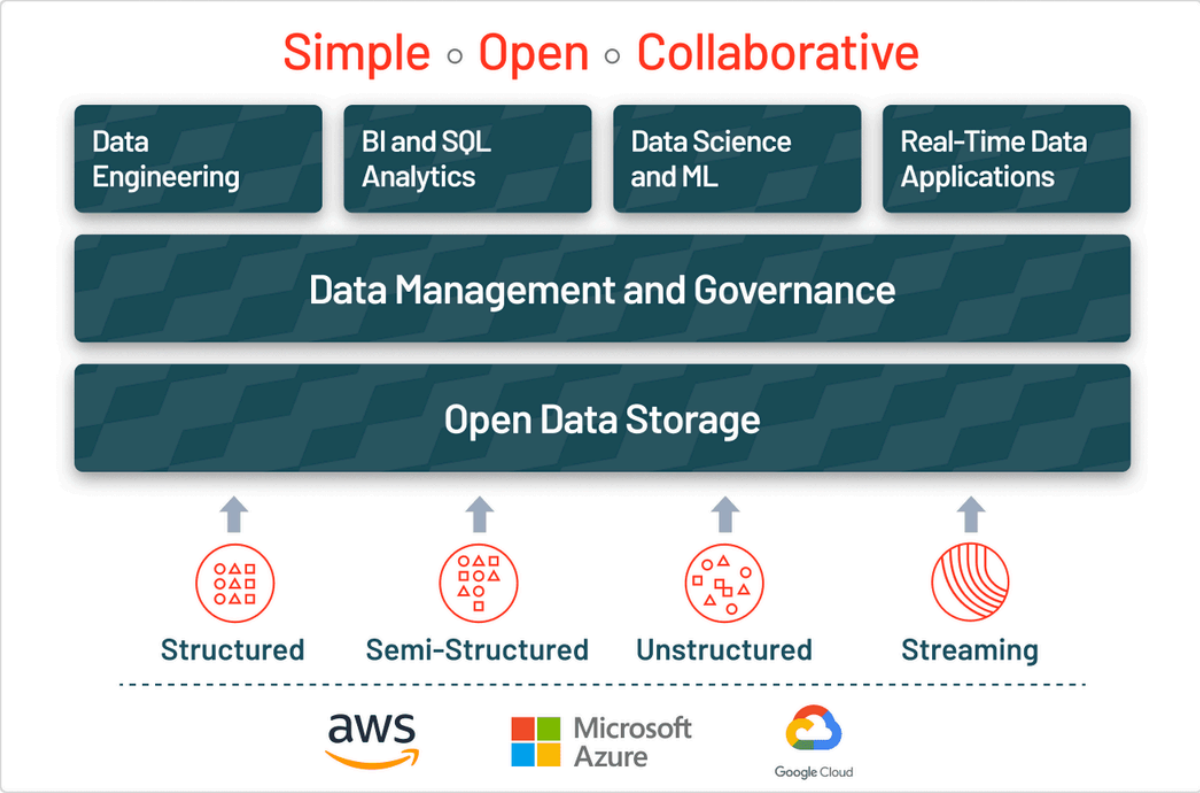
Building a lakehouse with Delta Lake

To build a successful lakehouse, organizations have turned to Delta Lake, an open format data management and governance layer that combines the best of both data lakes and data warehouses. Across industries, enterprises are leveraging Delta Lake to power collaboration by providing a reliable, single source of truth. By delivering quality, reliability, security and performance on your data lake — for both streaming and batch operations — Delta Lake eliminates data silos and makes analytics accessible across the enterprise. With Delta Lake, customers can build a cost-efficient, highly scalable lakehouse that eliminates data silos and provides self-serving analytics to end users.



Data lakes vs. data lakehouses vs. data warehouses

	Data lake	Data lakehouse	Data warehouse
Types of data	All types: Structured data, semi-structured data, unstructured (raw) data	All types: Structured data, semi-structured data, unstructured (raw) data	Structured data only
Cost	$	$	$$$
Format	Open format	Open format	Closed, proprietary format

Scalability	Scales to hold any amount of data at low cost, regardless of type	Scales to hold any amount of data at low cost, regardless of type	Scaling up becomes exponentially more expensive due to vendor costs
Intended users	Limited: Data scientists	Unified: Data analysts, data scientists, machine learning engineers	Limited: Data analysts
Reliability	Low quality, data swamp	High quality, reliable data	High quality, reliable data
Ease of use	Difficult: Exploring large amounts of raw data can be difficult without tools to organize and catalog the data	Simple: Provides simplicity and structure of a data warehouse with the broader use cases of a data lake	Simple: Structure of a data warehouse enables users to quickly and easily access data for reporting and analytics
Performance	Poor	High	High

Lakehouse best practices

Use the data lake as a landing zone for all of your data

Save all of your data into your data lake without transforming or aggregating it to preserve it for machine learning and data lineage purposes.

Mask data containing private information before it enters your data lake

Personally identifiable information (PII) must be pseudonymized in order to comply with GDPR and to ensure that it can be saved indefinitely.

Secure your data lake with role- and view-based access controls

Adding view-based ACLs (access control levels) enables more precise tuning and control over the security of your data lake than role-based controls alone.

Build reliability and performance into your data lake by using Delta Lake

The nature of big data has made it difficult to offer the same level of reliability and performance available with databases until now. Delta Lake brings these important features to data lakes.

Catalog the data in your data lake

Use data catalog and metadata management tools at the point of ingestion to enable self-service data science and analytics.

Feature	Data Lake	Data Warehouse
Schema	Schema-on-read	Schema-on-write
Data Types	Unstructured, semi-structured	Structured
Use Cases	ML, data science, streaming	BI, dashboards, reporting
Storage Cost	Lower	Higher
Performance	Variable	High for SQL workloads

Data lake vs. data warehouse

Now you know what a data lake is, why it matters, and how it's used across a variety of organizations. But what's the difference between a data lake and a

data warehouse? And when is it appropriate to use one over the other?

While data lakes and data warehouses are similar in that they both store and process data, each have their own specialties, and therefore their own use cases. That's why it's common for an enterprise-level organization to include a data lake and a data warehouse in their analytics ecosystem. Both repositories work together to form a secure, end-to-end system for storage, processing, and faster time to insight.

A data lake captures both relational and non-relational data from a variety of sources—business applications, mobile apps, IoT devices, social media, or streaming—without having to define the structure or schema of the data until it is read. Schema-on-read ensures that any type of data can be stored in its raw form. As a result, data lakes can hold a wide variety of data types, from structured to semi-structured to unstructured, at any scale. Their flexible and scalable nature make them essential for performing complex forms of data analysis using different types of compute processing tools like Apache Spark or Azure Machine Learning.

By contrast, a data warehouse is relational in nature. The structure or schema is modeled or predefined by business and product requirements that are curated, conformed, and optimized for SQL query operations. While a data lake holds data of all structure types, including raw and unprocessed data, a data warehouse stores data that has been treated and transformed with a specific purpose in mind, which can then be used to source analytic or operational reporting. This makes data warehouses ideal for producing more standardized forms of BI analysis, or for serving a business use case that has already been defined.

	Data lake	Data warehouse
Type	Structured, semi-structured, unstructured	Structured
	Relational, non-relational	Relational
Schema	Schema on read	Schema on write

Format	Raw, unfiltered	Processed, vetted
Sources	Big data, IoT, social media, streaming data	Application, business, transactional data, batch reporting
Scalability	Easy to scale at a low cost	Difficult and expensive to scale
Users	Data scientists, data engineers	Data warehouse professionals, business analysts
Use cases	Machine learning, predictive analytics, real-time analytics	Core reporting, BI

Data lake vs. data lakehouse

Now you know the difference between a data lake vs. a data warehouse. But what's the difference between a data lake and a data lakehouse? And is it necessary to have both?

Despite its many advantages, a traditional data lake is not without its drawbacks. Because data lakes can accommodate all types of data from all kinds of sources, issues related to quality control, data corruption, and improper partitioning can occur. A poorly managed data lake not only tarnishes data integrity, but it can also lead to bottlenecks, slow performance, and security risks.

That's where the data lakehouse comes into play. A data lakehouse is an open standards-based storage solution that is multifaceted in nature. It can address the needs of data scientists and engineers who conduct deep data analysis and processing, as well as the needs of traditional data warehouse professionals who curate and publish data for business intelligence and reporting purposes. The beauty of the lakehouse is that each workload can seamlessly operate on top of the data lake without having to duplicate the data into another structurally predefined database. This ensures that

everyone is working on the most up-to-date data, while also reducing redundancies.

Data lakehouses address the challenges of traditional data lakes by adding a Delta Lake storage layer directly on top of the cloud data lake. The storage layer provides a flexible analytic architecture that can handle ACID (atomicity, consistency, isolation, and durability) transactions for data reliability, streaming integrations, and advanced features like data versioning and schema enforcement. This allows for a range of analytic activity over the lake, all without compromising core data consistency. While the necessity of a lakehouse depends on how complex your needs are, its flexibility and range make it an optimal solution for many enterprise orgs.

	Data lake	Data lakehouse
Type	Structured, semi-structured, unstructured	Structured, semi-structured, unstructured
	Relational, non-relational	Relational, non-relational
Schema	Schema on read	Schema on read, schema on write
Format	Raw, unfiltered, processed, curated	Raw, unfiltered, processed, curated, delta format files
Sources	Big data, IoT, social media, streaming data	Big data, IoT, social media, streaming data, application, business, transactional data, batch reporting
Scalability	Easy to scale at a low cost	Easy to scale at a low cost

Users	Data scientists	Business analysts, data engineers, data scientists
Use cases	Machine learning, predictive analytics	Core reporting, BI, machine learning, predictive analytics